

## Project Initialization and Planning Phase

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|----------------------|---------------------------------------------------------------------|
| <b>Date</b>          | 29 September 2026                                                   |
| <b>Team ID</b>       | SWTID-2026-1374                                                     |
| <b>Project Title</b> | Script-Controlled ACL – Restrict Record Access Based on Field Value |
| <b>Maximum Marks</b> | 5 Marks                                                             |

### Project Proposal (Proposed Solution) report

The proposal report aims to protect student records in an institution using role-based and field-based access control in ServiceNow. It ensures that only authorized users can access EEE branch records, while administrators retain complete access to all records.

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| <b>Project Overview</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                |
| Objective                | To implement Access Control Lists (ACLs) in ServiceNow that control access to student records based on the user's role and the value of the Branch field.                                                                                                                                                                                                                                                      |
| Scope                    | A custom Institution Details table, custom roles (bb1–bb4), and READ, CREATE, WRITE and DELETE ACLs. The READ ACL uses a script and a branch-based condition to restrict EEE branch records, while administrators keep complete access.                                                                                                                                                                        |
| <b>Problem Statement</b> |                                                                                                                                                                                                                                                                                                                                                                                                                |
| Description              | Student information in an institution must be protected from unauthorized access. Different users need different levels of access such as viewing, creating, editing and deleting records, so a role-based and field-based access control mechanism is required.                                                                                                                                               |
| Impact                   | Solving this protects sensitive student information, provides role-based access, restricts records by branch, and prevents unauthorized creation, modification and deletion.                                                                                                                                                                                                                                   |
| <b>Proposed Solution</b> |                                                                                                                                                                                                                                                                                                                                                                                                                |
| Approach                 | Create a custom Institution Details table, create multiple roles and assign them to users, then configure ACLs for Read, Create, Write and Delete. The READ ACL uses a script and a branch-based condition to restrict access to EEE records.                                                                                                                                                                  |
| Key Features             | <ul style="list-style-type: none"><li>• Role-based access with roles bb1 (Read), bb2 (Create), bb3 (Write/Edit) and bb4 (Delete).</li><li>• READ ACL script that checks the Branch field so EEE records are available only to the authorized role.</li><li>• Complete access to all records for administrators.</li><li>• Prevention of unauthorized creation, modification and deletion of records.</li></ul> |

**Resource Requirements**

| Resource Type           | Description                             | Specification/Allocation                                                                                         |
|-------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Hardware</b>         |                                         |                                                                                                                  |
| Computing Resources     | CPU/GPU specifications, number of cores | Cloud-hosted ServiceNow instance (no local GPU needed)                                                           |
| Memory                  | RAM specifications                      | Managed by the ServiceNow platform                                                                               |
| Storage                 | Disk space for data, models, and logs   | ServiceNow instance database storage                                                                             |
| <b>Software</b>         |                                         |                                                                                                                  |
| Frameworks              | Platform                                | ServiceNow (Personal Developer Instance)                                                                         |
| Libraries               | Additional libraries                    | ServiceNow server-side JavaScript (Glide API) for the ACL script                                                 |
| Development Environment | IDE                                     | ServiceNow web UI – Tables, Roles, Users and Access Control (ACL) script editor                                  |
| <b>Data</b>             |                                         |                                                                                                                  |
| Data                    | Source, size, format                    | Student records entered in u_institution_details; sample records for ECE, EEE and CSE branches; ServiceNow table |